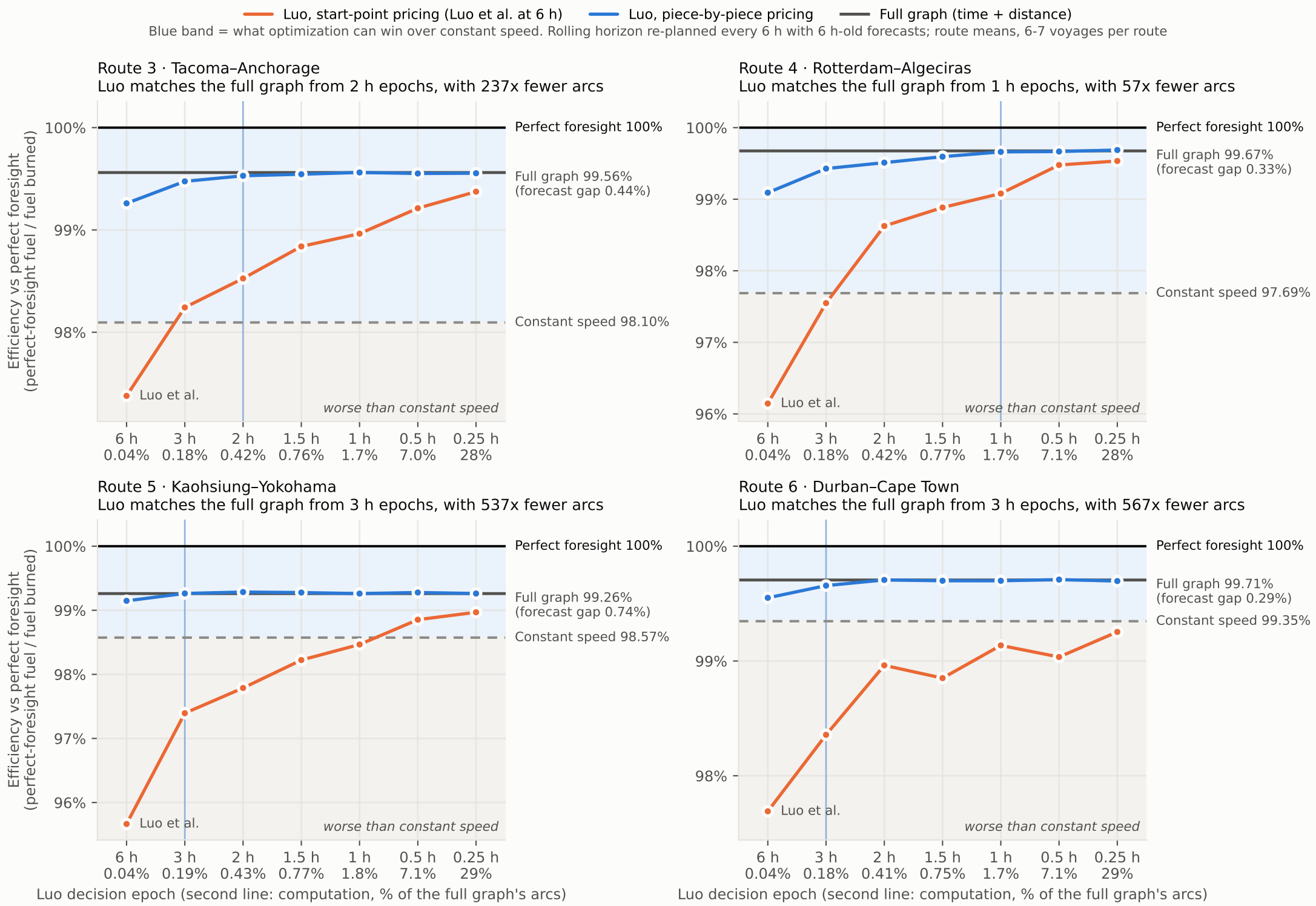
Luo, start-point pricing (Luo et al. at 6 h)
Luo, piece-by-piece pricing
Full graph (time + distance)
Blue band = what optimization can win over constant speed. Rolling horizon re-planned every 6 h with 6 h-old forecasts; route means, 6-7 voyages per route
Route 3 · Tacoma–Anchorage
Luo matches the full graph from 2 h epochs, with 237x fewer arcs
Route 4 · Rotterdam–Algeciras
Luo matches the full graph from 1 h epochs, with 57x fewer arcs
Route 5 · Kaohsiung–Yokohama
Luo matches the full graph from 3 h epochs, with 537x fewer arcs
Route 6 · Durban–Cape Town
Luo matches the full graph from 3 h epochs, with 567x fewer arcs
Efficiency vs perfect foresight
(perfect-foresight fuel / fuel burned)
100%
99%
98%
97%
96%
Perfect foresight 100%
Full graph 99.56%
(forecast gap 0.44%)
Constant speed 98.10%
Full graph 99.67%
(forecast gap 0.33%)
Constant speed 97.69%
Full graph 99.26%
(forecast gap 0.74%)
Constant speed 98.57%
Full graph 99.71%
(forecast gap 0.29%)
Constant speed 99.35%
Luo et al.
worse than constant speed
6 h 0.04%
3 h 0.18%
2 h 0.42%
1.5 h 0.76%
1 h 1.7%
0.5 h 7.0%
0.25 h 28%
6 h 0.04%
3 h 0.18%
2 h 0.42%
1.5 h 0.77%
1 h 1.7%
0.5 h 7.1%
0.25 h 28%
6 h 0.04%
3 h 0.19%
2 h 0.43%
1.5 h 0.77%
1 h 1.8%
0.5 h 7.1%
0.25 h 29%
6 h 0.04%
3 h 0.18%
2 h 0.41%
1.5 h 0.75%
1 h 1.7%
0.5 h 7.1%
0.25 h 29%
Luo decision epoch (second line: computation, % of the full graph's arcs)
Luo decision epoch (second line: computation, % of the full graph's arcs)